

A Novel CNN Regression based Reversible Data Hiding for Adaptive Embedding without Overhead

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A Novel CNN Regression based Reversible Data Hiding for Adaptive Embedding without Overhead

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Abstract

To deal with limited embedding capacity, Seucrity and Privacy concerns due to communication overhead, and issues related to faithful recovery of cover with traditional CNN methods, a novel framework is developed that leverages a fully reversible Convolutional Neural Network (CNN), based on an adapted iUnet architecture, to enable high-payload, overhead-free data hiding. The invertible CNN proactively reshapes the cover image's histogram through learnable transformations, creating zero-frequency bins to preemptively nullify overflow and underflow errors that cause overhead. This process simultaneously boosts histogram peaks, facilitating robust multi-bit-per-pixel embedding using either single or dual-peak schemes. Crucially, the same CNN is applied in reverse post-embedding to significantly enhance the marked image's visual quality and obscure statistical artifacts, thereby improving both imperceptibility and security. Evaluated on the ImageNet dataset, the proposed method produces zero overhead and demonstrates superior performance over state-of-the-art techniques. It achieves an average Peak Signal-to-Noise Ratio (PSNR) of 40.26 dB and a Structural Similarity Index (SSIM) of 0.9821 for its single-peak scheme, while guaranteeing complete recovery of the hidden data and the original cover image. This work establishes an adaptive paradigm that resolves the trade-off between capacity and overhead, advancing the state-of-the-art in secure and efficient RDH.

Keywords: Reversible Data Hiding, Histogram Shifting, U-Net, iUnets

1. Introduction

Data hiding is the sophisticated discipline that uses statistical properties of digital media to covertly embed ancillary information. This technique involves the deliberate manipulation of the host medium—be it an image, video, text file, or audio stream—to clandestinely sequester a data payload. The principal objective is to integrate the hidden information in such a manner that it remains perceptually and statistically inconspicuous, thereby ensuring its sequestered and confidential passage within the cover medium.

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Reversible Data Hiding (RDH)[3] is a technique that has been explored extensively for more than two decades. It is a technique in which data is concealed in an image. This data is then recovered by the receiver. Further, the original image is recovered completely. This technique has found great use in the military[29, 20, 7], medical[31, 32], and judicial fields[34, 33], where the complete recovery of cover media and the hidden message is paramount.

Kumar et al.[4] classified Reversible Data Hiding methods into two domains spatial domain and the encrypted domain. The spatial domain methods include methods like lossless expansion[5], correlation expansion[6, 20], histogram shifting[29, 8, 9, 10, 11], pixel value ordering[26], prediction error expansion[27, 28] and interpolation. Encrypted domain methods can be classified further into methods that Reserve Room Before Encryption[12, 13] and methods that Reserve Room After Encryption[14, 15]. Kumar et al.[9] use both of these techniques for bidirectional histogram shifting, reserving room before encryption using bidirectional histogram shifting and after encryption by embedding the overhead in the encrypted image.

The U-Net was first introduced by Ronneberger et al.[30] for medical image segmentation. It used connections between each encoder and decoder level to preserve the learnings from the feature maps at each level. However, this model was not reversible as it used non-invertible downsampling.

The IUNets model proposed by Etmann et al.[1] solved the memory constraints in traditional U-Net by using backtracking from [2, 16, 17]. This method requires that the input be divided into 2 parts before processing it in the model, and this was achieved in these methods by making the first downsampling non-invertible. We adapted this model for our use by introducing invertible downsampling in the first step itself.

The use of a U-Net-based architecture, like our adapted iUnet, allows for desired image modification while keeping the characteristics of the feature maps at each level using skip connections.

The various Histogram-Shifting-based methods explored in the past two decades faced several challenges. These methods suffered from a fixed embedding capacity due to the fixed number of peak value pixels in an image, which is extremely low for effective embedding. They also produced overhead bits as a result of embedding, which was either embedded in the image itself or transmitted over a separate channel. This either reduced the embedding capacity or compromised security. Traditional Histogram Shifting-based methods produce a "step-like" distribution in the region where data is embedded, making it easier for attackers to identify the region in which data is hidden.

To overcome these issues, we propose the following framework:

- We propose a CNN based generalized *framework* to create the necessary features and space for embedding data, thereby making our method completely independent of the features of the cover media. This model is made completely reversible and memory efficient by using invertible downsampling.
- We propose the use of the Reversible model to eliminate the *overflow* and *underflow* created during standard histogram shifting by using learnable gamma correction and clamping to redistribute the pixel value histogram of the cover image.
- We propose an *overhead* elimination strategy by requiring no side-channel communication for auxiliary infor-

mation, such as a location map, by using an adaptive histogram shifting method.

- We propose an invertible marked *cover enhancement* scheme which increases imperceptibility of the marked cover, hence making the scheme more robust.
- We propose a *security enhancement* scheme by using the reversible model to retrieveably obscure the pixel value histogram where the data is hidden. This is done by redistributing the pixel values again to remove the step-like artifact created in the histogram due to embedding. In addition to this, the lack of side channel communication makes the method more robust.
- We propose a *high embedding capacity* multi-peak multi-bit histogram shifting scheme, which is made possible due to the elimination of underflow and overflow, which previously would have made such a large-scale embedding impractical. Additionally, the enhancement scheme mitigates the quality degradation caused by high payload embedding, making it unobtrusive.
- The aforementioned features result in better performance metrics than the state-of-the-art methods [9, 10, 11]. These results were observed while experiments were performed on various kinds of cover images.

The remaining paper is categorized as follows: Section 2 outlines the review of recent related papers and highlights the research gaps. In Section 3, the core design of the proposed method is elaborated. The proposed method is compared with the state-of-the-art methods and results are summarized in Section 4. Concluding remarks and future directions are sketched in Section 5.

2. Related Works and Research Gaps

This section reviews the studies that influenced the design of our proposed method. We'll begin with a look at traditional Reversible Data Hiding (RDH) methods, focusing on Histogram Shifting (HS) to explain the core concepts and their limitations. Then, we will cover the key developments in neural network architecture, from the U-Net to fully invertible models, that made a new approach to RDH possible. Finally, by analyzing a recent, related technique, we will highlight the specific research gaps that this paper addresses.

Traditional Reversible Data Hiding and Histogram Shifting

Reversible Data Hiding (RDH) is a specialized subfield of data hiding where secret information is embedded into a cover medium, such as a digital image, with the critical guarantee that the original cover can be perfectly and losslessly restored after data extraction. This property of faithful recovery is non-negotiable in sensitive domains like medical imaging and digital forensics. RDH techniques are generally categorized into two main approaches: those that operate in the plain domain (on original pixel values) and those that work within an encrypted domain for enhanced privacy.

Among the foundational plain-domain techniques, Histogram Shifting (HS) emerged as a prominent method. The seminal work by Ni et al. established the core principle of manipulating an image’s histogram to create space for data by identifying the ”peak point” (the most frequent pixel value) and shifting a portion of the histogram to create a vacant bin adjacent to it. Early research focused on refining this concept to increase the payload. Fallahpour and Sedaaghi adapted the technique to operate on image blocks, while others like Lee et al. and Lin et al. moved from using the raw image histogram to that of the difference image to leverage the sharper, more centralized distributions for higher embedding potential. This trend was extended to multi-dimensional histograms to better exploit correlations in color images, as demonstrated by Chang et al.’s work.

While these methods successfully increased raw capacity, they were often constrained by the need to manage auxiliary data created during the shifting process. In response, techniques were developed to handle this more efficiently, such as Tai et al.’s use of a binary tree structure. Other innovative approaches sought to proactively reshape the histogram before embedding. For instance, Ying et al. proposed a method to enhance image contrast by merging bins, while Xiao et al. and Li et al. investigated using multiple peak and zero bin pairs for expansion.

These traditional HS methods were all fundamentally constrained by a few key research gaps. The most significant was the generation of ”overhead” required for reversibility, which reduced the net payload and introduced potential security vulnerabilities. Additionally, the embedding capacity remained intrinsically linked to the statistical properties of the cover image, limiting adaptability. These limitations established the need for more advanced techniques.

A Review of the Foundational CNN models

The CNN architecture used in this paper is closely related to the U-Net proposed by Ronneberger et al. for biomedical image segmentation. The U-Net’s distinctive ”U-shaped” design consists of a contracting path that captures context through downsampling and an expansive path that enables precise localization through upsampling. A key innovation was its use of skip connections, which concatenate feature maps from the contracting path to the corresponding layers in the expansive path. This allows the network to combine high-resolution spatial information with deep, contextual features, a crucial element for precise image-to-image tasks. However, the original U-Net architecture was not reversible due to its use of non-invertible operations like max-pooling.

The concept of invertibility in neural networks was explored extensively in the context of generative models. The work on Non-linear Independent Components Estimation (NICE) by Dinh et al. introduced the idea of using a series of simple, invertible transformations called ”coupling layers” to model complex data distributions. This was further developed into the Real NVP (Real-valued Non-Volume Preserving) framework by Dinh et al. (2016), which demonstrated how to build powerful, high-dimensional generative models with tractable log-likelihoods by stacking these affine coupling layers.

The iUNet adapts the U-Net’s multi-scale structure but replaces its standard components with fully invertible counterparts. This includes replacing standard convolutions with reversible blocks based on coupling layers and,

crucially, replacing non-invertible pooling and upsampling operations with learnable, invertible up- and downsampling methods. This full invertibility not only makes the architecture suitable for applications like normalizing flows but also allows for memory-efficient backpropagation, which is critical for training on large-scale, high-dimensional data such as 3D medical images. The architecture in the present paper builds directly on this foundation, adapting the iUnet by introducing a fully invertible downsampling mechanism from the very first step, thereby creating a completely reversible and memory-efficient model suitable for the unique demands of Reversible Data Hiding.

Analysis of Bidirectional and Multistage Histogram Shifting

A closely related and recent state-of-the-art technique is the bidirectional and multistage embedding method proposed by Kumar et al.. This work directly confronts the challenge of managing overhead to achieve high capacity and security. The core innovation is a two-pronged approach: first, it increases the initial payload by using bidirectional histogram shifting, which exploits peaks on both the left and right sides of the main histogram peak for data embedding. Second, it proposes a novel multistage process to handle the overhead generated during this spatial-domain embedding. Instead of transmitting the overhead through a separate, vulnerable channel or embedding it back into the plain cover image at the cost of capacity, this method embeds the overhead into a secondary, encrypted version of the cover image.

This hybrid plain-domain/encrypted-domain approach successfully contains all necessary information within the transmitted image, thereby enhancing security. However, this technique also highlights a persistent research gap. The method does not eliminate the generation of overhead in the spatial domain; rather, it introduces a complex, multi-stage process to manage and conceal it. This shifts the problem to the encrypted domain. This reliance on a secondary embedding stage introduces procedural complexity.

Research Gaps

The evolution of Histogram Shifting (HS) techniques reveals a persistent set of challenges that have driven innovation in the field of Reversible Data Hiding (RDH). Traditional HS methods, while foundational, are fundamentally constrained by an intrinsic trade-off between embedding capacity and the management of overhead. This overhead, generated by pixel value overflow and underflow, must be stored to ensure perfect reversibility, which either reduces the net data payload or compromises security by requiring a separate communication channel.

Furthermore, the embedding capacity of these methods is inherently dependent on the natural statistical properties of the cover image, such as the height of its histogram peaks. This limits their adaptability and performance on images with flat or uniform histograms. Even advanced techniques that employ multistage embedding to conceal overhead within the encrypted domain do not solve the root cause of overhead generation.

This highlights a clear need for a spatial domain RDH framework that can achieve high-capacity, multi-bit-per-pixel embedding without being constrained by the cover image’s statistics and without generating any overhead. The proposed method directly addresses this gap by employing an invertible neural network to proactively and

reversibly reshape the image histogram, completely eliminating the cause of overflow and underflow errors and thus eradicating the need for overhead communication. This approach decouples the embedding capacity from the image's original properties, establishing an adaptive paradigm that resolves the trade-off between capacity and overhead.

The following section details the architecture and operational pipeline of the proposed method, demonstrating how it achieves high-capacity, secure, and completely overhead-free reversible data hiding.

3. Proposed Framework for Secret Message Communication

The main bottleneck with Histogram Shifting-based Reversible Data Hiding is that it produces an overhead when the data is hidden. This overhead is either sent through a separate channel, compromising security, or embedded within the image itself, impacting the data hiding capacity and quality of the image. For this, we propose a reversible CNN-based architecture that can create bins with frequency zero in the region that would otherwise fall into the overhead.

The proposed method, as illustrated in Fig. 1 is comprised of four stages. At the transmission end, a cover image is pre-processed by squeezing its channels and using a CNN to create zero-frequency values, which are then consolidated via a leftward histogram shift to create embedding space. The secret message is embedded by modifying pixel values at the histogram peaks, and another CNN enhances the quality of the final marked cover image. At

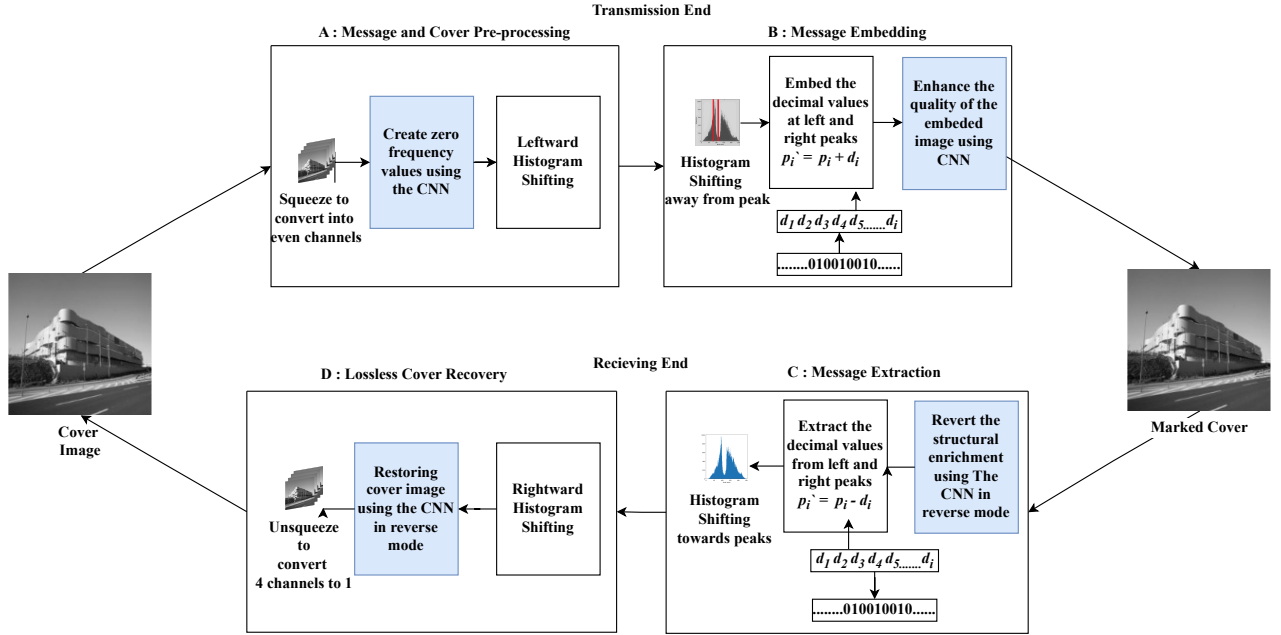


Figure 1: Overview of the Proposed Four-Stage Pipeline for Adaptive Reversible Data Hiding using a CNN to Create Zero Frequency Histogram Bins in the Desired Region and using Bidirectional Histogram Shifting for Embedding and the reverse process for Message Extraction and Cover Recovery.

the receiving end, the process is inverted: the message is first extracted by reverting the CNN enhancement and reading the histogram peak modifications. Subsequently, the original image is losslessly recovered through a rightward histogram shift, an inverse CNN operation, and an unsqueeze transformation to restore the original channels.

3.1. Core Design and Key Concepts of Proposed Data Embedding Method

The input image $\mathcal{I}$ was passed through the squeeze function, $\psi(\mathcal{I})$, to get the output with 4 channels and half the dimensions ($\mathcal{I}_1$), which was passed through the reversible CNN. The reversible CNN was used to redistribute the histogram of the image to create the zero frequency bins in the region $(0, 2 \times 2^n - 1)$, where $\mathbf{n}$ is the number of bits to be embedded per pixel, to eliminate the overhead created during histogram shifting, creating image ($\mathcal{I}_2$). The process also “boosts” the peak values in the histogram as a byproduct of the zero-frequency bin creation.

In this paper, we use the squeeze operation as both an invertible downsampling as well a preprocessing. The encoder applies a squeeze operation (ψ), at each level that reduces the spatial dimension while increasing the channel depth. Each squeeze operation transforms a tensor from shape $(\mathcal{B}, \mathcal{C}, \mathcal{H}, \mathcal{W})$ where $\mathcal{B}$ is the batch size, $\mathcal{C}$ the channel width, and the dimensions of each channel are given by $\mathcal{H}$ and $\mathcal{W}$, to $(\mathcal{B}, 4\mathcal{C}, \mathcal{H}/2, \mathcal{W}/2)$ as shown in figure 2. This transformation is completely invertible through a corresponding unsqueeze operation (ψ^{-1}).

$$\psi : \mathbb{R}^{\mathcal{C} \times \mathcal{H} \times \mathcal{W}} \rightarrow \mathbb{R}^{4\mathcal{C} \times \mathcal{H}/2 \times \mathcal{W}/2} \quad (1)$$

$$\psi^{-1} : \mathbb{R}^{4\mathcal{C} \times \mathcal{H}/2 \times \mathcal{W}/2} \rightarrow \mathbb{R}^{\mathcal{C} \times \mathcal{H} \times \mathcal{W}} \quad (2)$$

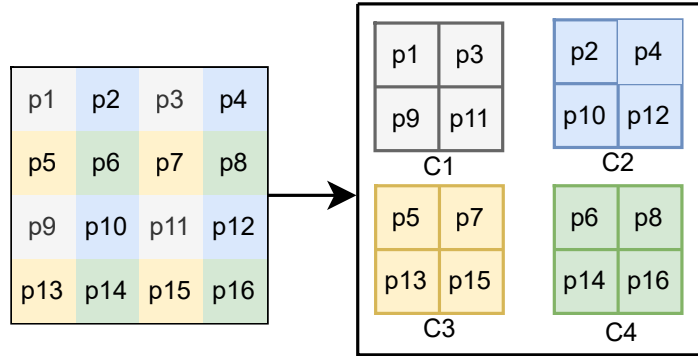


Figure 2: Squeeze operation for Pre-processing and Downsampling.

This model uses a modified U-Net architecture for the image-to-image regression task with complete reversibility. Similar to a traditional U-Net[30], this model also uses skip connections that connect the encoder and decoder at the same level, which preserves the spatial information at that level, which is otherwise lost in downsampling, while

also ensuring invertibility at each level. Both the Reversible Block and the modified U-Net require input tensors in which the number of channels is even, as they split the input for processing.

Overview of the Proposed CNN Architecture

The additive coupling rule, based on prior work such as NICE[17] and RealNVP[16], forms the basis of reversibility of the network in the Reversible Block which divides an input $\mathcal{X}$, into two parts ($\mathcal{X}_1$) and ($\mathcal{X}_2$) on which convolutions $\mathcal{F}()$ and $\mathcal{G}()$ are applied producing output ($\mathcal{Y}_1$) and ($\mathcal{Y}_2$) which are combined to give us output $\mathcal{Y}$.

$$\mathcal{Y}_1 = \mathcal{X}_1 + \mathcal{F}(\mathcal{X}_2) \quad (3)$$

$$\mathcal{Y}_2 = \mathcal{X}_2 + \mathcal{G}(\mathcal{Y}_1) \quad (4)$$

Here the functions $\mathcal{F}()$ and $\mathcal{G}()$ consist of a convolution layer with block size $\mathcal{C} \times 3 \times 3$ followed by a bounded $\tanh()$ activation rather than ReLU, which keeps the output within the desired range and removes the outliers which would result in the predicted value exceeding 255. This is followed by another convolutional layer with the same kernel size. A Dropout layer is added which temporarily deactivates some neurons during training, thus preventing overfitting.

$$\begin{aligned} \mathcal{F}(), \mathcal{G}() : \text{Conv2D}(X) \rightarrow (\tanh(X) \rightarrow \text{Conv2D}(X))^2 \rightarrow \\ (\tanh(X) \rightarrow \text{Dropout}() \rightarrow \text{Conv2D} \end{aligned} \quad (5)$$

The functions $\mathcal{F}()$ and $\mathcal{G}()$ being convolutions with a lossy activation function are not inherently reversible but the transformations are made reversible by the inverse of Equations 3 and 4.

$$\mathcal{X}_2 = \mathcal{Y}_2 - \mathcal{G}(\mathcal{Y}_1) \quad (6)$$

$$\mathcal{X}_1 = \mathcal{Y}_1 - \mathcal{F}(\mathcal{X}_2) \quad (7)$$

The Dropout layer [18] is only used during training and does not affect the reversibility of the model during its operation.

At each encoding level, the network splits the input tensor into 2 equal chunks, one of which is used for a skip connection to the decoder, while the other is used for the main processing. The portioned input goes through a squeeze block in order to increase channel depth and decrease spatial dimensions for more efficient learning. Then this downsampled tensor is processed by the Reversible Block, which carries out the convolutions.

Similarly, at each decoder level, the input is first passed through a Reversible block. This output is then unsqueezed in order to get an upsampled image which has a smaller number of channels and expanded spatial dimensions. This unsqueezed tensor is concatenated with the tensor from the skip connection with the encoder at

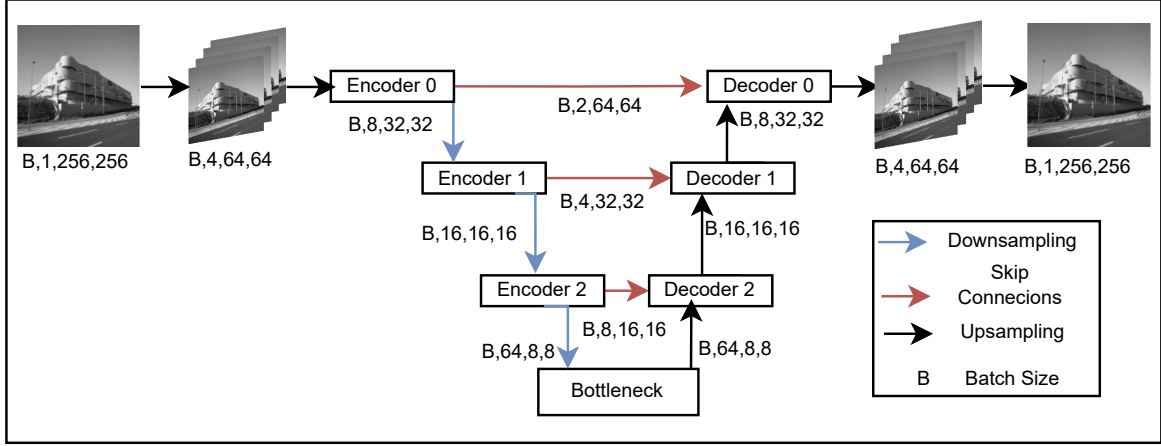


Figure 3: The adapted iUNets-based model for creation of Zero Frequency Bins in the Desired Range

for Overhead-free Histogram Shifting based Reversible Data Hiding with Squeeze Pre-processing.

the same level. The bottleneck, which consists of only a single Reversible block, is placed at the transition between the last encoder and the first decoder. This placement in the most information-dense and mathematically well-controlled point in the network allows it to make complex, information-rich transformations to the network while ensuring complete mathematical invertibility.

For the inverse operation, the architecture remains the same with just the reversible block changed to its reverse mode.

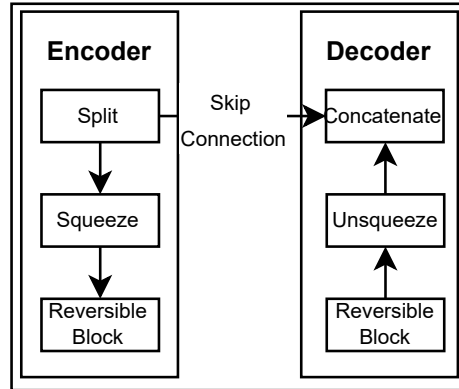


Figure 4: The components of an encoder-decoder pair, which form the basis for the reversibility and target image creation for overhead-free Reversible Data Hiding.

Creation of Target Images The creation of target images for the model training involves two steps: gamma correction and clipping. Gamma correction [19] is applied pixel-wise to increase the intensity of each pixel in the lighter intensity range. Using a gamma value $\gamma = 0.8$, we boost the darker, i.e., low intensity, pixel values, causing

more pixels to move towards high intensities. This results in a non-linear shift of the histogram peaks towards the right. This results in compression of pixel values in the darker regions, redistributing them towards the lighter side of the histogram.

The clipping operation “raises” the value of any pixel that falls below a threshold to the threshold value. This was done to remove the overhead as shown later.

$$\mathcal{I}_{target} = \max(255 \cdot (\mathcal{I}_{original}/255)^\gamma, 255) \quad (8)$$

Training parameters and loss functions The network parameters ω are optimized by back-propagation using the Adam optimizer[35] with learning rate $\alpha = 10^{-4}$ and weight decay $\lambda = 10^{-5}$.

The model uses two kinds of losses for effective training and to make sure no pixel value falls below the threshold value, a MSE loss between the model output ($\hat{I}$) and the corresponding target image $\mathcal{I}_{target}$ and a specially designed forbidden region loss that makes sure that any pixel value below the threshold κ incurs a linear penalty.

$$\mathcal{L}_{MSE} = \frac{1}{NHW} \sum_{j=1}^N \sum_{u=1}^H \sum_{v=1}^W \left(\widehat{I^{(j)}}(u, v) - I_{target}^{(j)}(u, v) \right)^2 \quad (9)$$

$$\mathcal{L}_{low} = \frac{1}{NHW} \sum_{j=1}^N \sum_{u=1}^H \sum_{v=1}^W \max \left(w - \widehat{I^{(j)}}(u, v), 0 \right) \quad (10)$$

$$\mathcal{L}_{high} = \frac{1}{NHW} \sum_{j=1}^N \sum_{u=1}^H \sum_{v=1}^W \max \left(\widehat{I^{(j)}}(u, v) - 255, 0 \right) \quad (11)$$

where N is the batch size and $H \times W$ are the dimensions of the image. The model essentially applies learnable gamma correction, which shifts the peak of the image towards the lighter intensity region. This transformation is reversed during the inverse operation of the network, shifting the peaks back towards the lighter region.

Message pre-processing

We take a list of secret data and a chunk size $\mathbf{n}$ as input and convert the binary data into decimal values by grouping them into chunks of size $\mathbf{n}$. If the length of the data list is not divisible by $\mathbf{n}$, it is padded with zeros. Then, we iterate over the padded data list, converting each chunk of binary data into a decimal value and storing it in a new list. The choice of the chunk size $\mathbf{n}$ depends on the number of shifts made in the histogram. For embedding a chunk size of $\mathbf{n}$, we need to shift the histogram by $(2^n - 1)$ places.

3.2. Message Embedding

The histogram of an image is the mapping of pixel intensities to the frequency distribution of the occurrence of each pixel in the image. The peak of the image is the pixel intensity with the maximum frequency. We refer to it as the main peak, p_{main} .

$$p_{main} = \arg \max (0 \leq p \leq 255) \quad (12)$$

For our method, we also use the left peak(p_{left}) and right peak(p_{right}) of the image, which are the pixel values with the maximum frequency that are less than and greater than the main peak, respectively.

$$p_{left} = \arg \max (0 \leq p \leq p_{main}) \quad (13)$$

$$p_{right} = \arg \max (p_{main} \leq p \leq 255) \quad (14)$$

Embedding in Left and Right Peak

First, the reversible CNN is used to create the bins with frequency zero in the first 15 bins of the lighter region by using the model trained with the value of $\kappa = 15$.

Since this method requires that there be at least $(2^n - 1)$ zero frequency pixel values at the dark as well as light regions of the image, we redistribute the zeroes frequency bins created using the reversible CNN to create an image($\mathcal{I}_3$) suitable for bi-directional histogram shifting without overhead.

$$\mathcal{I}_3 = \mathcal{I}_2 - 2^n - 1 \quad (15)$$

In order to embed the data, we need to create gaps in the histogram of size $(2^n - 1)$ by shifting the left peaks leftwards and the right peak rightwards, creating image ($\mathcal{I}_4$). This transformation usually produces an overhead but due to our CNN-based approach, the need for overhead management is eliminated.

$$I_4(u, v) = \begin{cases} \mathcal{I}_3(u, v) - (2^n - 1) & \text{if } \mathcal{I}_3(u, v) \leq p_{left} \\ \mathcal{I}_3(u, v) + (2^n - 1) & \text{if } \mathcal{I}_3(u, v) \geq p_{right} \\ \mathcal{I}_3(u, v) & \text{otherwise} \end{cases} \quad (16)$$

Since the histogram is shifted, the peak values of this new image are also shifted as follows,

$$p'_{left} = p_{left} - (2^n - 1) \quad (17)$$

$$p'_{right} = p_{right} - (2^n - 1) \quad (18)$$

Data is embedded at the gaps that we created at the peaks through histogram shifting by adding the data decimal value (d_i) to the left peak or subtracting the decimal value (d_i) from the right peak to “fill up” the gaps

created in the histogram. This gives us image ($\mathcal{I}_5$).

$$\mathcal{I}_5(u, v) = \begin{cases} \mathcal{I}_4(u, v) - d_i & \text{if } I_4(u, v) = p'_{right} \\ \mathcal{I}_4(u, v) + d_i & \text{if } I_4(u, v) = p'_{left} \\ \mathcal{I}_4(u, v) & \text{if } I_4(u, v) \neq p'_{left}, p'_{right} \end{cases} \quad (19)$$

Algorithm 1 Single Peak Embedding using CNN-based Reversible Data Hiding

Require: Cover image I , Secret data S , Chunk size n , Reversible CNN model

Ensure: Marked image I_{marked}

```

1:  $I_2 \leftarrow \text{CNN\_forward}(I)$  {Pre-processing with reversible CNN}
2:  $H \leftarrow \text{histogram}(I_2)$ 
3:  $p_{\text{main}} \leftarrow \arg \max(H)$  {Find main histogram peak}
4:  $S_{\text{groups}} \leftarrow \text{group\_bits}(S, n)$  {Convert secret bits to decimal}
5:  $I_3 \leftarrow \text{shift\_left}(I_2, p_{\text{main}}, 2^n - 1)$  {Create embedding gap}
6: for each pixel  $(u, v)$  in  $I_3$  do
7:   if  $I_3(u, v) = p_{\text{main}} - (2^n - 1)$  then
8:      $I_3(u, v) \leftarrow I_3(u, v) + d_i$  where  $d_i \in S_{\text{groups}}$ 
9:   end if
10: end for
11:  $I_{\text{marked}} \leftarrow \text{CNN\_reverse}(I_3)$  {Post-process to enhance quality}
12: return  $I_{\text{marked}}$ 

```

Embedding in Main Peak

In order to embed the data, we need to create a gap in the histogram of size $(2^n - 1)$ by shifting the main peak leftwards, creating image ($\mathcal{I}_4$). This transformation usually produces an overhead, but due to our CNN-based approach, the need for overhead management is eliminated.

$$I_4(u, v) = \begin{cases} I_2(u, v) - (2^n - 1) & \text{if } I_2(u, v) \leq p_{\text{main}} \\ I_2(u, v) & \text{otherwise} \end{cases} \quad (20)$$

Since the histogram is shifted, the peak value of this new image are also shifted as follows,

$$p'_{\text{main}} = p_{\text{main}} - (2^n - 1) \quad (21)$$

Data is embedded at the gap that we created at the peak through histogram shifting by adding the data decimal value (d_i) to the main peak to “fill up” the gap created in the histogram. This gives us image $\mathcal{I}_5$.

$$\mathcal{I}_5(u, v) = \begin{cases} \mathcal{I}_4(u, v) + d_i & \text{if } I_4(u, v) = p'_{main} \\ \mathcal{I}_4(u, v) & \text{if } I_4(u, v) \neq p'_{main} \end{cases} \quad (22)$$

$(\mathcal{I}_5)$ is passed through the CNN in reverse mode, which increases the peak-signal-to-noise-ratio (PSNR) between the marked cover and the original cover image. This is done by performing the inverse of the CNN operation by shifting the peaks back towards the left and introducing low intensity pixels back into the image, resulting in Marked Image, $(\mathcal{I}_{marked})$.

Algorithm 2 Multi-Peak Embedding using Bidirectional Histogram Shifting

Require: Cover image I , Secret data S , Chunk size n , Reversible CNN model

Ensure: Marked image I_{marked}

```

1:  $I_2 \leftarrow \text{CNN\_forward}(I)$  {Pre-processing with reversible CNN}
2:  $H \leftarrow \text{histogram}(I_2)$ 
3:  $p_{\text{left}} \leftarrow \text{find\_left\_peak}(H)$ 
4:  $p_{\text{right}} \leftarrow \text{find\_right\_peak}(H)$ 
5:  $S_{\text{groups}} \leftarrow \text{group\_bits}(S, n)$  {Convert secret bits to decimal}
6:  $I_3 \leftarrow \text{shift\_left}(I_2, p_{\text{left}}, 2^n - 1)$  {Create left embedding gap}
7:  $I_3 \leftarrow \text{shift\_right}(I_3, p_{\text{right}}, 2^n - 1)$  {Create right embedding gap}
8: for each pixel  $(u, v)$  in  $I_3$  do
9:   if  $I_3(u, v) = p_{\text{left}} - (2^n - 1)$  then
10:     $I_3(u, v) \leftarrow I_3(u, v) + d_i$  where  $d_i \in S_{\text{groups}}$ 
11:   else if  $I_3(u, v) = p_{\text{right}} + (2^n - 1)$  then
12:     $I_3(u, v) \leftarrow I_3(u, v) - d_i$  where  $d_i \in S_{\text{groups}}$ 
13:   end if
14: end for
15:  $I_{marked} \leftarrow \text{CNN\_reverse}(I_3)$  {Post-process to enhance quality}
16: return  $I_{marked}$ 

```

3.3. Message Extraction

The Marked Cover $(\mathcal{I}_{marked})$ is first passed through the CNN to reverse the quality enhancements made during the embedding process. This image $(\mathcal{I}_5)$ is then used for data extraction through histogram shifting.

Extracting from Left and Right peaks

The secret message in decimal form (d_i) is extracted from the histogram peaks in the image $(\mathcal{I}_5)$ by subtracting the left peak from the pixel values in the gap created near the left peak at the time of embedding. Similarly, the data is extracted from the right peak by adding the right peak value to the pixels with values in the range, which was deemed a histogram gap during the embedding.

$$d_i = \begin{cases} \mathcal{I}_5(u, v) - p'_{left} & \text{if } p'_{left} \leq I_4(u, v) \leq p'_{left} + (2^n - 1) \\ \mathcal{I}_5(u, v) + p'_{right} & \text{if } p'_{right} - (2^n - 1) \leq I_4(u, v) \leq p'_{right} \end{cases} \quad (23)$$

Once the data is recovered, we use the data in order to recover the gaps that were created in the histogram during embedding recovering image ($\mathcal{I}_4$)

$$\mathcal{I}_4(u, v) = \begin{cases} \mathcal{I}_5(u, v) - d_i & \text{if } p'_{left} \leq I_4(u, v) \leq p'_{left} + (2^n - 1) \\ \mathcal{I}_5(u, v) + d_i & \text{if } p'_{right} - (2^n - 1) \leq I_4(u, v) \leq p'_{right} \\ \mathcal{I}_5(u, v) & \text{otherwise} \end{cases} \quad (24)$$

After data extraction, the histogram bins must be shifted back to their original positions to perfectly recover the cover image. For every bin in $[0, p'_{left}]$, add $2^n - 1$ to move frequencies back towards the left peak. Similarly, for every bin in $[p'_{right}, 255]$, subtract $(2^n - 1)$ to move frequencies back towards the right peak, recovering image ($\mathcal{I}_3$)

$$\mathcal{I}_3(u, v) = \begin{cases} \mathcal{I}_4(u, v) + (2^n - 1) & \text{if } I_4(u, v) \leq p'_{left} \\ \mathcal{I}_4(u, v) - (2^n - 1) & \text{if } I_4(u, v) \geq p'_{right} \end{cases} \quad (25)$$

The histogram, which was shifted towards the left to create zero value bins on both sides of the histogram, is shifted back towards the right to restore image ($\mathcal{I}_2$).

$$\mathcal{I}_2 = \mathcal{I}_3 + 2^n - 1 \quad (26)$$

Algorithm 3 Multi-Peak Extraction using Bidirectional Histogram Shifting

Require: Marked image I_{marked} , Chunk size n , Reversible CNN model

Ensure: Extracted secret data S , Recovered cover image I

```
1:  $I_5 \leftarrow \text{CNN\_forward}(I_{\text{marked}})$ 
2:  $H \leftarrow \text{histogram}(I_5)$ 
3:  $p'_{\text{left}} \leftarrow \text{find\_left\_peak}(H)$ 
4:  $p'_{\text{right}} \leftarrow \text{find\_right\_peak}(H)$ 
5: Initialize empty list  $S_{\text{groups}}$ 
6: for each pixel  $(u, v)$  in  $I_5$  do
7:   if  $I_5(u, v)$  in range  $[p'_{\text{left}}, p'_{\text{left}} + (2^n - 1)]$  then
8:      $d_i \leftarrow I_5(u, v) - p'_{\text{left}}$ 
9:     Append  $d_i$  to  $S_{\text{groups}}$ 
10:     $I_4(u, v) \leftarrow I_5(u, v) - d_i$ 
11:   else if  $I_5(u, v)$  in range  $[p'_{\text{right}} - (2^n - 1), p'_{\text{right}}]$  then
12:      $d_i \leftarrow p'_{\text{right}} - I_5(u, v)$ 
13:     Append  $d_i$  to  $S_{\text{groups}}$ 
14:      $I_4(u, v) \leftarrow I_5(u, v) + d_i$ 
15:   else
16:      $I_4(u, v) \leftarrow I_5(u, v)$ 
17:   end if
18: end for
19:  $I_3 \leftarrow \text{shift\_right}(I_4, p'_{\text{left}}, 2^n - 1)$ 
20:  $I_3 \leftarrow \text{shift\_left}(I_3, p'_{\text{right}}, 2^n - 1)$ 
21:  $I_2 \leftarrow \text{shift\_right}(I_3, p'_{\text{left}}, 2^n - 1)$  {Restore histogram bins}
22:  $I_1 \leftarrow \text{CNN\_reverse\_squeeze}(I_2)$ 
23:  $I \leftarrow \text{CNN\_reverse}(I_1)$  {Recover original cover image}
24:  $S \leftarrow \text{binary\_from\_decimal}(S_{\text{groups}}, n)$ 
25: return  $S, I$ 
```

Extracting the Secret Message From the Main Peak

Once the data is recovered, we use the data in order to recover the gaps that were created in the histogram during embedding recovering image ($\mathcal{I}_4$)

$$\mathcal{I}_4(u, v) = \begin{cases} \mathcal{I}_5(u, v) - d_i & \text{if } p'_{\text{main}} \leq I_4(u, v) \leq p'_{\text{main}} \\ & + (2^n - 1) \\ \mathcal{I}_5(u, v) & \text{otherwise} \end{cases} \quad (27)$$

After data extraction, the histogram bins must be shifted back to their original positions to perfectly recover the cover image. For every bin in $[0, p'_{\text{main}}]$, add $2^n - 1$ to move frequencies back towards the left peak, recovering image ($\mathcal{I}_2$)

$$\mathcal{I}_2(u, v) = \mathcal{I}_4(u, v) + (2^n - 1) \text{if } I_4(u, v) \leq p'_{\text{main}} \quad (28)$$

The secret message in decimal form (d_i) is extracted from the histogram peak in the image ($\mathcal{I}_5$) by subtracting

the main peak from the pixel values in the gap created near the main peak at the time of embedding.

$$d_i = \mathcal{I}_5(u, v) - p'_{main} \text{ if } p'_{main} \leq I_4(u, v) \leq p'_{main} + (2^n - 1) \quad (29)$$

Algorithm 4 Single Peak Extraction using Reversible Data Hiding

Require: Marked image I_{marked} , Chunk size n , Reversible CNN model

Ensure: Extracted secret data S , Recovered cover image I

```

1:  $I_5 \leftarrow \text{CNN\_forward}(I_{\text{marked}})$  {Reverse quality enhancements}
2:  $H \leftarrow \text{histogram}(I_5)$ 
3:  $p'_{\text{main}} \leftarrow \arg \max(H)$  {Find shifted main peak}
4: Initialize empty list  $S_{\text{groups}}$ 
5: for each pixel  $(u, v)$  in  $I_5$  do
6:   if  $I_5(u, v)$  in range  $[p'_{\text{main}}, p'_{\text{main}} + (2^n - 1)]$  then
7:      $d_i \leftarrow I_5(u, v) - p'_{\text{main}}$ 
8:     Append  $d_i$  to  $S_{\text{groups}}$ 
9:      $I_4(u, v) \leftarrow I_5(u, v) - d_i$ 
10:  else
11:     $I_4(u, v) \leftarrow I_5(u, v)$ 
12:  end if
13: end for
14:  $I_2 \leftarrow \text{shift\_right}(I_4, p'_{\text{main}}, 2^n - 1)$  {Restore histogram bins}
15:  $I_1 \leftarrow \text{CNN\_reverse\_squeeze}(I_2)$  {Inverse preprocessing}
16:  $I \leftarrow \text{CNN\_reverse}(I_1)$  {Recover original cover image}
17:  $S \leftarrow \text{binary\_from\_decimal}(S_{\text{groups}}, n)$  {Convert to binary secret}
18: return  $S, I$ 

```

3.4. Cover Recovery

The image ($\mathcal{I}_2$) was then squeezed to convert it into a 4-channel image in order to facilitate processing by the CNN in reverse mode to restore the original image.

$$\psi(\mathcal{I}_2) = \mathcal{I}_1 \quad (30)$$

The cover image($\mathcal{I}$) is finally recovered by passing the image ($\mathcal{I}_1$) through the CNN in reverse mode which reverses all of the transformations and returns the cover image($\mathcal{I}$).

The following section provides a comprehensive evaluation of this framework to validate its performance and demonstrate its superiority over existing methods.

4. Experimental Validation

The performance of the proposed method was evaluated using a subset of the ImageNet dataset. The evaluation demonstrates that the proposed framework achieves accurate data retrieval and image reconstruction at the

receiver's end.

4.1. Experimental Design

The experiments were conducted using 100 images selected from the ImageNet test dataset. For the purpose of this paper, these images were converted to grayscale and resized to a uniform dimension of 256×256 to serve as cover images.

The performance of the proposed algorithm was assessed based on three key metrics:

- **Embedding Capacity (EC):** This metric measures the maximum amount of secret data that can be hidden within the cover image, typically expressed in bits per pixel (bpp) or total bits.
- **Peak Signal-to-Noise Ratio (PSNR):** PSNR is used to quantify the quality of the marked cover image relative to the original. It is measured in decibels (dB), with higher values indicating better image quality. It is derived from the Mean-Squared Error (MSE).
 - Mean-Squared Error aims to compare between pixel values of original $\mathcal{I}(u, v)$ and pixel value of marked cover $\mathcal{I}'(u, v)$ image of size $\mathcal{M} \times \mathcal{N}$, and is determined by the square of the differences of the original and marked cover image. Mathematically, it is defined as,

$$MSE = \frac{1}{M \times N} \sum_{i=1}^M \sum_{j=1}^N [\mathcal{I}(u, v) - \mathcal{I}'(u, v)]^2 \quad (31)$$

The value of MSE should be minimum and, in general, falls between $[0, \infty)$.

- PSNR is defined as:

$$PSNR = 10 \cdot \log_{10} \left(\frac{255^2}{MSE} \right) \quad (32)$$

- **Structural Similarity Index** is a quality assessment metric of marked cover image, which determines the similarity between the original cover $\mathcal{I}(u, v)$ and marked cover $\mathcal{I}'(u, v)$ image. The value of SSIM lies between $[0, 1]$ and it should be very close to one. This is determined as,

$$SSIM = \left(\frac{2\mu_{\mathcal{I}}\mu_{\mathcal{I}'} + c_1}{2\mu_{\mathcal{I}}^2\mu_{\mathcal{I}'}^2 + c_1} \right) \left(\frac{2\sigma_{\mathcal{I}\mathcal{I}'} + c_2}{\sigma_{\mathcal{I}}^2\sigma_{\mathcal{I}'}^2 + c_2} \right) \quad (33)$$

where $\mu_{\mathcal{I}}, \mu_{\mathcal{I}'}$ and $\sigma_{\mathcal{I}}^2, \sigma_{\mathcal{I}'}^2$ are the averages and variances of original and marked cover images, respectively. $\sigma_{\mathcal{I}\mathcal{I}'}$ is the covariance of the original and marked image, c_1 and c_2 are the regularization constants.

The proposed method's results were compared against several state-of-the-art techniques, including those by Fallahpour et al. [10], Wang et al. [11], Ni et al. [29], and Kumar et al. [9].

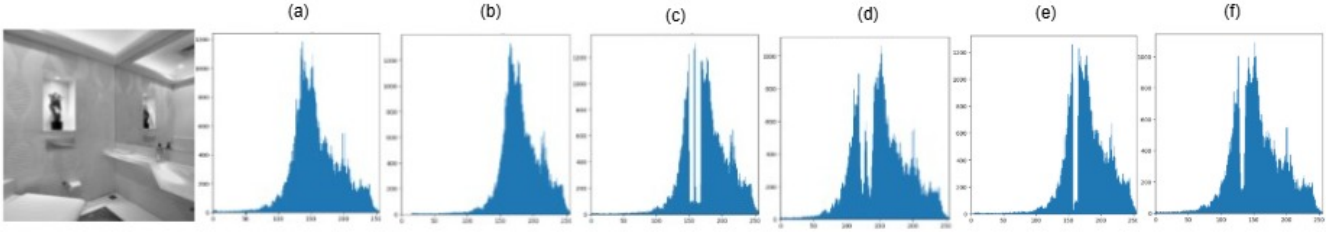


Figure 5: ILSVRC2013_train_00090004 (Image 1) (a) Histogram of the image (b) Histogram after creating zero bins using the reversible CNN (c) Histogram after shifting both the peaks and embedding the data ($\kappa = 15$) (d) Histogram after enhancing the embedded image ($\kappa = 15$) using the CNN in reverse mode (e) Histogram after shifting the main peak and embedding the data ($\kappa = 7$) (f) Histogram after enhancing the embedded image ($\kappa = 7$) using the CNN in reverse mode

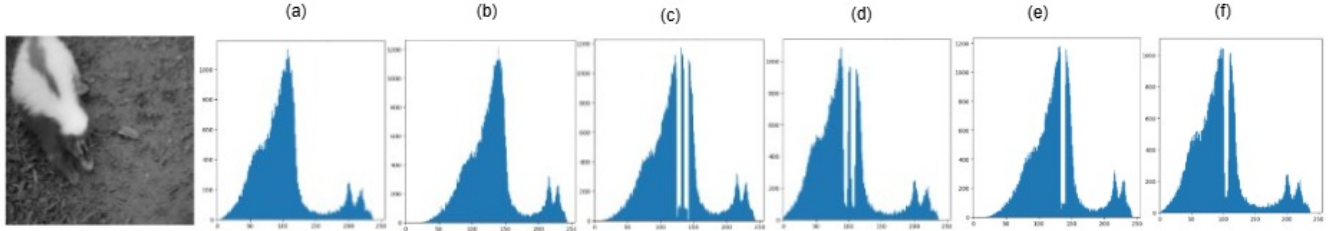


Figure 6: ILSVRC2017_test_00004364 (Image 2) (a) Histogram of the image (b) Histogram after creating zero bins using the reversible CNN (c) Histogram after shifting both the peaks and embedding the data ($\kappa = 15$) (d) Histogram after enhancing the embedded image ($\kappa = 15$) using the CNN in reverse mode (e) Histogram after shifting the main peak and embedding the data ($\kappa = 7$) (f) Histogram after enhancing the embedded image ($\kappa = 7$) using the CNN in reverse mode

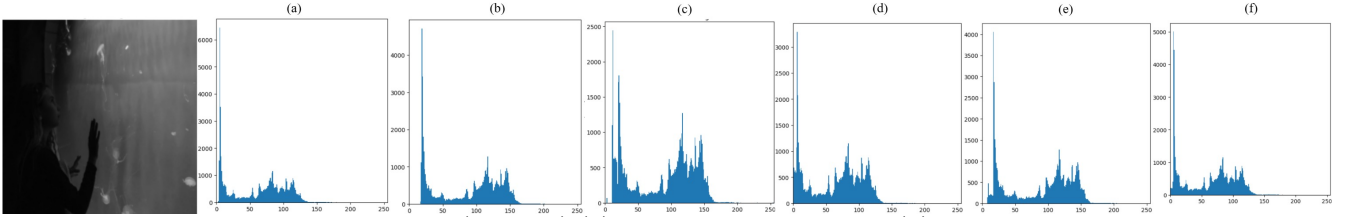


Figure 7: ILSVRC2017_test_00005242 (Image 3) (a) Histogram of the image (b) Histogram after creating zero bins using the reversible CNN (c) Histogram after shifting both the peaks and embedding the data ($\kappa = 15$) (d) Histogram after enhancing the embedded image ($\kappa = 15$) using the CNN in reverse mode (e) Histogram after shifting the main peak and embedding the data ($\kappa = 7$) (f) Histogram after enhancing the embedded image ($\kappa = 7$) using the CNN in reverse mode

4.2. Qualitative Analysis

The qualitative analysis shows the effectiveness of the proposed framework at each stage of the data hiding and recovery process. The visual evidence, presented through the series of histograms in Figures 5 through 12, provides a clear narrative that validates the method’s core claims.

The initial and most critical stage of the process is the proactive creation of zero-frequency bins using the reversible CNN. A visual comparison between the original image histograms (sub-figure ‘a’ in Figs. 5-12) and the post-CNN histograms (sub-figure ‘b’) demonstrates the effectiveness of this approach. In every case, regardless of the original histogram’s shape—whether it is wide and flat (Fig. 11a), narrow and peaked (Fig. 8a), or dark-biased

(Fig. 7a)—the CNN successfully reshapes the distribution by shifting pixel values to the right. This creates a distinct empty region on the left side, effectively carving out the necessary space to prevent underflow errors during the subsequent histogram shifting for embedding. This transformation highlights the method’s adaptive nature, as it is not dependent on the natural statistical properties of the cover image but rather creates the ideal conditions for embedding. A byproduct of this process is the ”boosting” of histogram peaks, which concentrates pixel values and facilitates a high embedding capacity.

Sub-figures ’c’ (dual-peak embedding) and ’e’ (single-peak embedding) clearly show the result of the data insertion step. The histograms in this phase exhibit the classic ”step-like” artifact characteristic of traditional histogram shifting methods. This plateau or flattened peak is a statistical anomaly that can make an image vulnerable to ste-

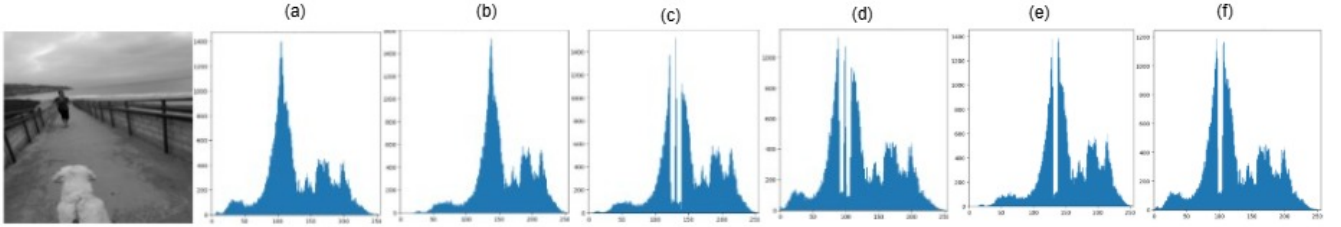


Figure 8: ILSVRC2017_test_00004869 (Image 4) (a) Histogram of the image (b) Histogram after creating zero bins using the reversible CNN (c)Histogram after shifting both the peaks and embedding the data ($\kappa = 15$)(d) Histogram after enhancing the embedded image ($\kappa = 15$) using the CNN in reverse mode (e)Histogram after shifting the main peak and embedding the data ($\kappa = 7$)(f) Histogram after enhancing the embedded image ($\kappa = 7$) using the CNN in reverse mode

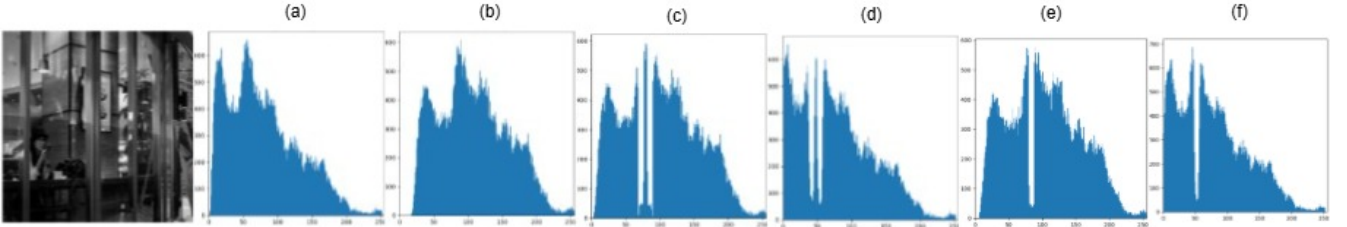


Figure 9: ILSVRC2017_test_00005123 (Image 5) (a) Histogram of the image (b) Histogram after creating zero bins using the reversible CNN (c)Histogram after shifting both the peaks and embedding the data ($\kappa = 15$)(d) Histogram after enhancing the embedded image ($\kappa = 15$) using the CNN in reverse mode (e)Histogram after shifting the main peak and embedding the data ($\kappa = 7$)(f) Histogram after enhancing the embedded image ($\kappa = 7$) using the CNN in reverse mode

ganalysis, as it signals that the histogram has been manipulated.

However, the most compelling qualitative result is the effect of the post-embedding enhancement, shown in the transition from sub-figures ’c’ to ’d’ and ’e’ to ’f’. In this stage, the same CNN is applied in reverse, and its impact is twofold. Firstly, it significantly enhances the visual quality of the marked image, which is quantitatively confirmed by the increase in PSNR and SSIM values reported in Table 2.

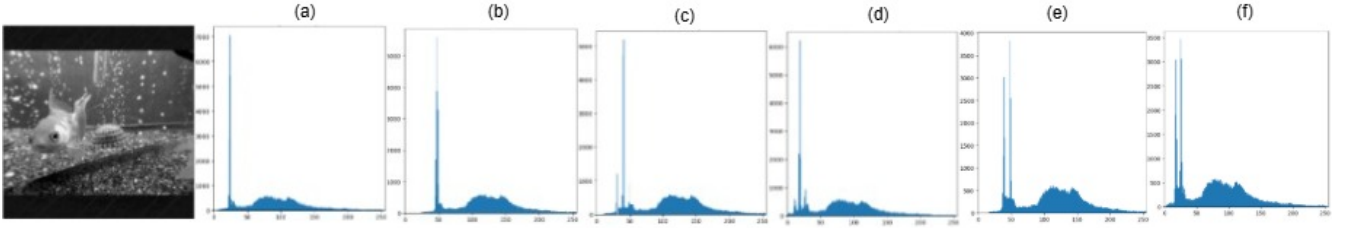


Figure 10: ILSVRC2017_test_00005135 (Image 6) (a) Histogram of the image (b) Histogram after creating zero bins using the reversible CNN (c)Histogram after shifting both the peaks and embedding the data ($\kappa = 15$) (d) Histogram after enhancing the embedded image ($\kappa = 15$) using the CNN in reverse mode (e)Histogram after shifting the main peak and embedding the data ($\kappa = 7$) (f) Histogram after enhancing the embedded image ($\kappa = 7$) using the CNN in reverse mode

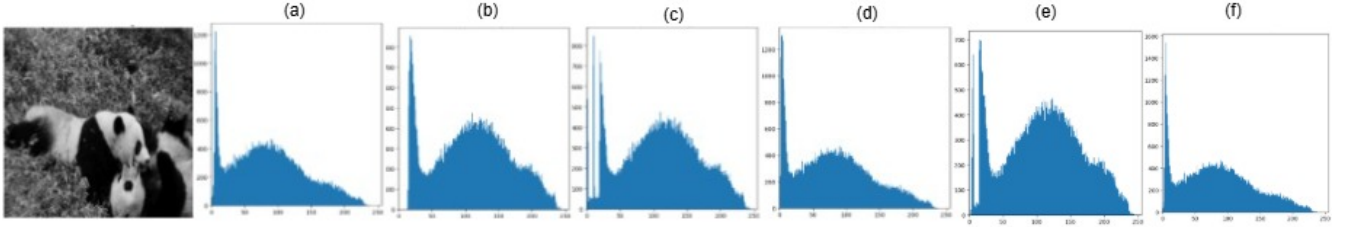


Figure 11: ILSVRC2017_test_00005243 (Image 7) (a) Histogram of the image (b) Histogram after creating zero bins using the reversible CNN (c)Histogram after shifting both the peaks and embedding the data ($\kappa = 15$) (d) Histogram after enhancing the embedded image ($\kappa = 15$) using the CNN in reverse mode (e)Histogram after shifting the main peak and embedding the data ($\kappa = 7$) (f) Histogram after enhancing the embedded image ($\kappa = 7$) using the CNN in reverse mode

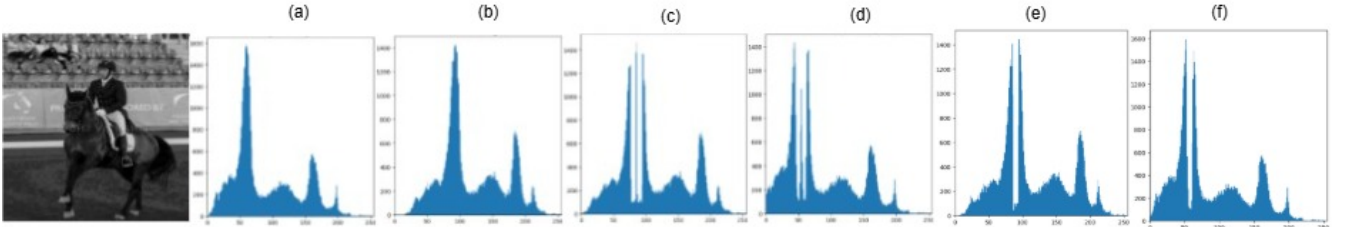


Figure 12: ILSVRC2017_test_00005341 (Image 8) (a) Histogram of the image (b) Histogram after creating zero bins using the reversible CNN (c)Histogram after shifting both the peaks and embedding the data ($\kappa = 15$) (d) Histogram after enhancing the embedded image ($\kappa = 15$) using the CNN in reverse mode (e)Histogram after shifting the main peak and embedding the data ($\kappa = 7$) (f) Histogram after enhancing the embedded image ($\kappa = 7$) using the CNN in reverse mode

Secondly, and crucially for security, this reverse pass obscures the step-like artifacts created during embedding. The histograms in sub-figures 'd' and 'f' appear much more natural and smooth, removing the obvious statistical evidence of data hiding. This act of "retrievably obscuring" the pixel value histogram is a powerful security feature that makes the marked image more resilient to detection.

4.3. Quantitative Analysis

A comprehensive quantitative analysis of the data presented in the tables validates the proposed framework's superior performance, highlighting its ability to resolve the persistent trade-offs between embedding capacity, image

Table 1: Performance metrics for different images using the proposed method

Images	Maximum capacity		PSNR		SSIM	
256×256	(bits)		(dB)			
	$\kappa = 15$	$\kappa = 7$	$\kappa = 15$	$\kappa = 7$	$\kappa = 15$	$\kappa = 7$
Image 1	7848	4032	29.63 dB	35.78 dB	0.9614	0.9860
Image 2	6534	3366	27.88 dB	33.52 dB	0.9579	0.9856
Image 3	13161	12648	43.81 dB	50.77 dB	0.9367	0.9912
Image 4	8409	4518	30.11 dB	36.00 dB	0.9614	0.9843
Image 5	3678	1947	31.28 dB	37.41 dB	0.9138	0.9817
Image 6	21810	14946	37.77 dB	41.89 dB	0.9651	0.9734
Image 7	4722	2103	43.88 dB	50.08 dB	0.9478	0.9762
Image 8	8658	4521	31.03 dB	36.64 dB	0.9375	0.9782
Average	8140.875	7224.5	34.47 dB	40.26 dB	0.9499	0.9821

quality, and overhead management in Reversible Data Hiding (RDH).

The internal performance of the proposed method, detailed in Table 1 and Table 2, demonstrates its flexibility and the critical role of its CNN-based enhancement. Table 1 evaluates two distinct schemes: a dual-peak approach ($\kappa = 15$) designed for maximum capacity and a single-peak approach ($\kappa = 7$) optimized for visual quality. The results show a well-managed trade-off; the dual-peak scheme achieves a high average embedding capacity of 8140.875 bits with an average Peak Signal-to-Noise Ratio (PSNR) of 34.47 dB, while the single-peak scheme delivers a superior average PSNR of 40.26 dB and an excellent Structural Similarity Index (SSIM) of 0.9821 for a slightly lower average capacity of 7224.5 bits. Furthermore, Table 2 quantifies the profound impact of the post-embedding enhancement step. On average, applying the CNN in reverse boosts the PSNR from a modest 20.77 dB to a strong 40.26 dB and increases the SSIM from 0.9314 to 0.9499. This confirms that the enhancement module is not an incremental improvement but a core component essential for achieving high-fidelity results.

Table 3 showcases the framework’s capability to completely eliminate overhead. The proposed method consistently produces zero overhead bits for every test image. This is a stark contrast to competing high-capacity methods like Kumar et al. [9], which generates a substantial average of 3538.125 overhead bits, and traditional methods like Ni et al. [29], which averages 140 bits and fails on at least one image due to excessive overhead. While the method by Wang et al. [11] also avoids overhead, it does so at the cost of a much lower embedding capacity, a limitation the proposed framework overcomes. This zero-overhead achievement is a fundamental quantitative advantage, directly enabling higher secure payloads.

Table 4 provides a direct comparison of embedding capacity, where the proposed method demonstrates its

Table 2: Comparison of Quality Metrics Before and After Enhancement For Eight Randomly Selected Images

Image 256×256	PSNR		SSIM	
	Before ment	Enhance- ment	Before ment	Enhance- ment
Image 1	22.25 dB	35.78 dB	0.9511	0.9614
Image 2	21.51 dB	33.52 dB	0.9453	0.9579
Image 3	26.41 dB	50.77 dB	0.8392	0.9367
Image 4	20.92 dB	36.00 dB	0.9474	0.9614
Image 5	20.13 dB	37.41 dB	0.9281	0.9138
Image 6	19.48 dB	41.89 dB	0.9025	0.9651
Image 7	19.42 dB	50.08 dB	0.8946	0.9478
Image 8	20.42 dB	36.64 dB	0.9146	0.9375
Average	20.77 dB	40.26 dB	0.9314	0.9499

Table 3: Comparison of the Number of Overhead Bits For Eight Randomly Selected Images

Image	Three-Bit Per Peak			One-Bit Per Peak	
	Proposed	[9]	[10]	[29]	[11]
Image 1	0	2380	8	144	0
Image 2	0	4632	8	16	0
Image 3	0	1024	8	720*	0
Image 4	0	2684	8	16	0
Image 5	0	1268	8	144	0
Image 6	0	10328	8	16	0
Image 7	0	3057	8	16	0
Image 8	0	2932	8	48	0
Average	0	3538.125	8	140	0

Table 4: Comparison of Embedding Capacity for Eight Randomly Selected Images

Image	Three-Bit Per Peak		One-Bit Per Peak			
	Proposed		[9]	[10]	[29]	[11]
	$\kappa = 15$	$\kappa = 7$				
Image 1	7848	4032	7148	1420	982	1126
Image 2	6534	3366		1047	1047	4632
Image 3	13161	12648	3258	756	-	506
Image 4	8409	4518	8328	1951	1404	1404
Image 5	3678	1947	3822	969	670	670
Image 6	21810	14946	19416	6614	6587	6603
Image 7	4722	2103	7225	2005	1408	1424
Image 8	8658	4521	9114	2079	1498	1546
Average	8140.875	7224.5	7867.875	2158.5	1942.29	1790.75

leadership. The dual-peak scheme’s average capacity of 8140.875 bits outperforms all other methods evaluated. It surpasses the comparable 3-bit-per-pixel approach of Kumar et al. [9] (7867.875 bits) and vastly exceeds the capacity of one-bit-per-pixel methods by margins ranging from 277% to over 354%. This quantitative dominance in payload, made possible by the overhead-free design, proves the framework’s effectiveness in maximizing the data hiding potential of an image.

Finally, Table 5 presents a nuanced comparison of image quality via PSNR. While low-capacity, one-bit methods achieve higher PSNR values (averaging from 51.66 dB to 68.24 dB) due to the minimal distortion they introduce, the most critical comparison is with other high-capacity techniques. Here, the proposed framework is vastly superior. Its single-peak scheme’s average PSNR of 40.26 dB completely eclipses the average of 9.04 dB achieved by Kumar et al. [9]. This demonstrates that for a similar or higher payload, the proposed method preserves image quality far more effectively, achieving an improvement of over 345%. Collectively, the quantitative data from these tables proves that the proposed method establishes a new state-of-the-art benchmark by achieving a higher embedding capacity than existing methods, completely eliminating overhead, and maintaining a much better balance between high payload and visual imperceptibility.

4.4. Experimental Result Summary

Based on the extensive evaluation, the proposed method demonstrates a superior balance between embedding capacity and image quality compared to existing high-capacity techniques. While one-bit-per-pixel approaches yield a higher quality marked image, they do so at the cost of a substantially lower embedding capacity. The proposed framework successfully achieves a higher embedding capacity than even the 3-bit-per-pixel approach of Kumar et

Table 5: Comparison of Peak Signal Noise Ratio For Eight Randomly Selected Images

Image	Three-Bit Per Peak		One-Bit Per Peak			
	Proposed		[9]	[10]	[29]	[11]
	$\kappa = 15$	$\kappa = 7$				
Image 1	29.63 dB	35.78 dB	8.82 dB	52.59 dB	53.52 dB	69.06 dB
Image 2	27.88 dB	33.52 db	8.83 dB	50.04 dB	-	69.31 dB
Image 3	43.81 dB	50.77 db	8.96 dB	51.45 dB	-	72.35 dB
Image 4	30.11 dB	36.00 dB	9.65 dB	50.81 dB	52.59 dB	68.05 dB
Image 5	31.28 dB	37.41 dB	8.75 dB	50.30 dB	51.62 dB	70.98 dB
Image 6	37.77 dB	41.89 dB	8.85 dB	58.46 dB	54.83 dB	61.04 dB
Image 7	43.88 dB	50.08 dB	9.31 dB	49.41 dB	-	67.60 dB
Image 8	31.03 dB	36.64 dB	9.17 dB	50.20 dB	51.88 dB	67.55 dB
Average	34.47 dB	40.26 dB	9.04 dB	51.66 dB	52.89 dB	68.24 dB

al., while simultaneously eliminating overhead and maintaining operations in the spatial domain for better quality preservation. The complete eradication of overhead bits by the reversible CNN is a significant advancement, directly enabling higher embedding payloads without compromising security or image quality.

5. Conclusion and Future Directions

By introducing a fully reversible CNN-based framework, the critical challenges of low capacity, security vulnerabilities, and cumbersome overhead in traditional histogram-shifting RDH have been successfully overcome. The primary contribution is the complete elimination of overhead bits, achieved by proactively creating zero-frequency bins with the invertible CNN before embedding. This method consistently produced no overhead bits on average, in stark contrast to the 3538.125 bits generated by comparable high-capacity techniques.

This overhead-free approach enables a significantly enhanced embedding capacity, making multi-bit-per-pixel embedding practical. Experimentally, the technique achieved an average capacity of 8140.875 bits, outperforming prior methods by margins ranging from 3.55% to over 300%. Furthermore, the framework boosts security by using the CNN in reverse post-embedding, which obscures "step-like" histogram artifacts and substantially improves the marked image's quality, increasing the average PSNR by 56.33%. Ultimately, this work establishes a new paradigm for RDH that balances a high payload with imperceptibility, delivering an average PSNR of 40.26 dB and SSIM of 0.9821 while outperforming similar high-capacity spatial-domain methods in quality by as much as 259%. In Future, this work can extend by exploring multilevel embedding across multiple histogram peaks to further increase data embedding capacity.

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Author Contributions :

Shubham Makhija- He worked on the conceptualization, Methodology, Validation and re- viewing of the manuscript.

Sanjay Kumar- He contributed to the methodology, formal analysis, data curation, re- source acquiring and writing original draft manuscript.

Gurjit Singh Walia- He worked on the formal analysis, resource acquiring, review, and editing the original manuscript.

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